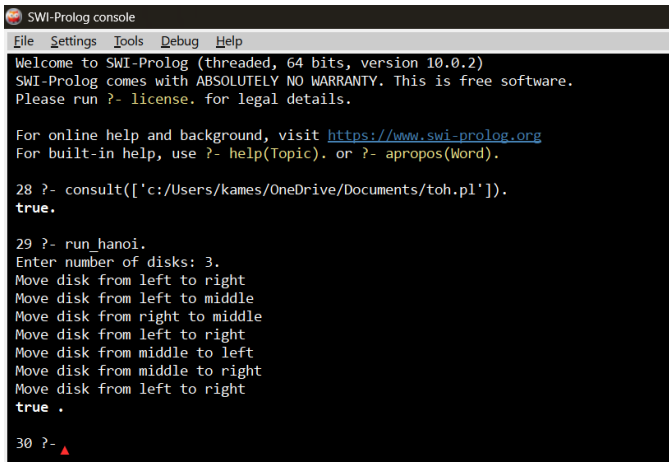


3-g Tower of Hanoi in Prolog

PROGRAM:

```
hanoi(1, Source, Dest, _) :-  
    format('Move disk from ~w to ~w~n', [Source, Dest]).  
  
hanoi(N, Source, Dest, Aux) :-  
    N > 1,  
    M is N - 1,  
    hanoi(M, Source, Aux, Dest),  
    hanoi(1, Source, Dest, Aux),  
    hanoi(M, Aux, Dest, Source).  
  
run_hanoi :-  
    write('Enter number of disks: '), read(N),  
    hanoi(N, left, right, middle).
```

OUTPUT:



```
SWI-Prolog console  
File Settings Tools Debug Help  
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)  
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.  
Please run ?- license. for legal details.  
  
For online help and background, visit https://www.swi-prolog.org  
For built-in help, use ?- help(Topic). or ?- apropos(Word).  
  
28 ?- consult(['c:/Users/kames/OneDrive/Documents/toh.pl']).  
true.  
  
29 ?- run_hanoi.  
Enter number of disks: 3.  
Move disk from left to right  
Move disk from left to middle  
Move disk from right to middle  
Move disk from left to right  
Move disk from middle to left  
Move disk from middle to right  
Move disk from left to right  
true .  
  
30 ?- ▲
```